APACHE (httpd) WEBSERVER

APACHE WEBSERVER

1. static web hosting
2. reverse proxy
3. load balancing
4. SSL termination
5. Virtual hosting (one Apache server hosts multiple websites)

HOSTING STATIC WEBSITE ON APACHE

1. yum install httpd -y
2. systemctl start httpd.service
3. systemctl enable httpd.service
4. cd /var/www/html/
5. vim index.html
   <h1> Hello Apache2! </h1>
6. systemctl restart httpd.service
7. http://public_ip

REVERSE PROXY ON APACHE

1. wget https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.120/bin/apache-tomcat-9.0.120.tar.gz
2. tar -xzf apache-tomcat-9.0.120.tar.gz
3. dnf install java-17-amazon-corretto-devel -y
4. cd apache-tomcat-9.0.120
5. cd bin
6. ./startup.sh
7. http://public_ip:8080

so suppose website is running on http://public_ip:8080

8. cd /etc/httpd/conf.d/
9. vim reverse-proxy.conf

```
<VirtualHost *:80>
   ServerName localhost
   ProxyPreserveHost On
   ProxyPass / http://127.0.0.1:8080/
   ProxyPassReverse / http://127.0.0.1:8080/
</VirtualHost>
```

10. apache2ctl configtest
11. systemctl restart httpd
12. http://public_ip

LOAD BALANCING ON APACHE

1. tomcat-server-1 (http://10.0.0.0:8080)
2. tomcat-server-2 (http://12.0.0.0:8080)
3. apache-lb-server (http://public_up)

4. vim /etc/httpd/conf.d/load-balancer.conf

```
<VirtualHost *:80>
   ServerName localhost
   ProxyPreserveHost On
   <Proxy "balancer://mycluster">
      BalancerMember "http://10.0.0.0:8080"
      BalancerMember "http://12.0.0.0:8080"
      ProxySet lbmethod=byrequests
   </Proxy>
   ProxyPass "/" "balancer://mycluster/"
   ProxyPassReverse "/" "balancer://mycluster/"
</VirtualHost>
```

5. apache2ctl configtest
6. systemctl restart httpd
7. http://public_ip
   { tomcat-1 / tomcat-2 }

LOAD BALANCER + DNS + SSL

1. tomcat-server-1 (http://10.0.0.0:8080)
2. tomcat-server-2 (http://12.0.0.0:8080)
3. apache-lb-server (http://public_ip)

Configure apache-lb-server

5.vim /etc/httpd/conf.d/load-balancer.conf

```
<VirtualHost *:80>
   ServerName mayurcbz.space
   ServerAlias www.mayurcbz.space
   ProxyPreserveHost On
   <Proxy "balancer://mycluster">
      BalancerMember "http://10.0.0.0:8080"
      BalancerMember "http://12.0.0.0:8080"
      ProxySet lbmethod=byrequests
   </Proxy>
   ProxyPass "/" "balancer://mycluster/"
   ProxyPassReverse "/" "balancer://mycluster/"
```

</VirtualHost>

6. systemctl restart httpd
7. Test the apache-lb-server
8. http://public_ip

9.Add the Apache-lb-server public-ip on the Hostinger A record

10.Test the Domain
http://mayurcbz.space

CONFIGURE SSL CERTIFICATES

1. sudo dnf install mod_ssl -y
2. sudo dnf install certbot python3-certbot-apache -y
3. certbot --apache
   Enter your email:
   Access the terms:
   Redirect: HTTP to HTTPS
4. certbot certificates
5. systemctl restart httpd
6. lsof -i :80
7. lsof -i :443

8. Test HTTPS
   https://mayurcbz.space

CERTIFICATE RENEVAL
   1. sudo certbot renew --dry-run

VIRTUAL HOSTING

1. dnf install httpd -y
2. systemctl start httpd
3. systemctl enable httpd
4. curl localhost

CREATE WEBSITE 1
1. mkdir -p /var/www/site1
2. vim /var/www/site1/index.html
   <h1> Welcome to Website-1 </h1>

CREATE WEBSITE-2
1. mkdir -p /var/www/site2
2. vim /var/www/site2/index.html
   <h1> Welcome to Website-2 </h1>

CREATE VIRTUAL HOST FOR WEBSITE 1

1.vim /etc/httpd/conf.d/site1.conf

```
<VirtualHost *:80>
   ServerName site1.mayurcbz.space
   DocumentRoot /var/www/site1
   <Directory /var/www/site1>
      AllowOverride None
      Require all granted
   </Directory>
</VirtualHost>
```

CREATE VIRTUAL HOST FOR WEBSITE 2

1.vim /etc/httpd/conf.d/site2.conf

```
<VirtualHost *:80>
   ServerName site2.mayurcbz.space
   DocumentRoot /var/www/site2
   <Directory /var/www/site2>
      AllowOverride None
      Require all granted
   </Directory>
</VirtualHost>
```

2.systemctl restart httpd

DNS CONFIGURATION ON HOSTINGER

1.add A record for site1
 Type: A
 Name: site1
 Value: APACHE-PUBLIC-IP
 TTL: 300

2.Add A record for site2
Type: A
Name: site2
Value: APACHE-PUBLIC-IP
TTL: 300

TEST THE WEBSITES

1. http://site1.mayurcbz.space
2. http://site2.mayurcbz.space
